

Trading Strategy Development Complete Reference Guide

Rule-Based, Machine Learning & Hybrid Approaches

Covering the complete lifecycle from hypothesis formation through live deployment.

Grounded in Lopez de Prado (2018), Chan (2009, 2013), Aronson (2006), and Pardo (2008).

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Complete 8-Phase Strategy Development Workflow

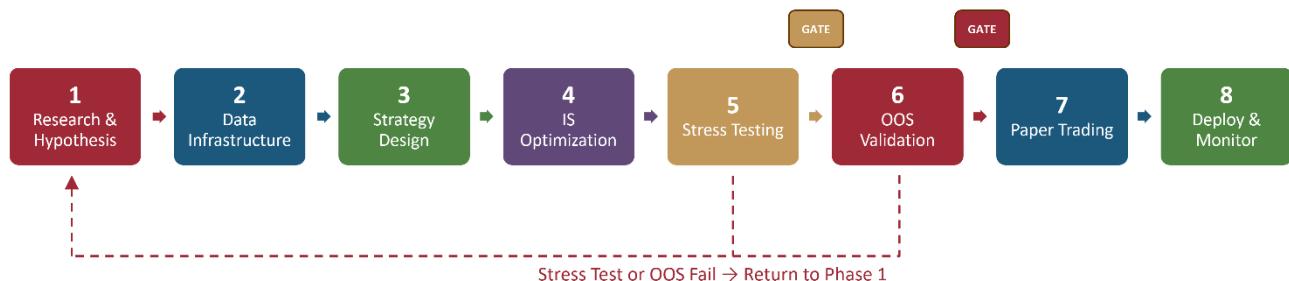


Figure 1. The complete 8-phase workflow. Each phase gates the next. Failure at stress testing (Phase 5) or OOS validation (Phase 6) returns the researcher to Phase 1 with a new hypothesis.

This document provides a comprehensive reference for building, testing, and deploying systematic trading strategies across three implementation paradigms: rule-based, machine learning, and hybrid. The eight-phase workflow is sequenced to reflect the logical dependencies between tasks. ML-specific concepts are integrated into the phases where they belong alongside their rule-based equivalents.

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PHASE 1: Research & Hypothesis Formation

Establishing a theoretically grounded, falsifiable edge before any data is examined

The most consequential discipline in quantitative strategy development is ensuring that a strategy's logic is theoretically motivated before any historical data is analyzed. Aronson (2006) and Lopez de Prado (2018) converge on this point: when a researcher examines data first and derives a hypothesis afterward, they are guaranteed to find patterns — markets contain enough noise that some apparent signal will always achieve statistical significance by chance. Phase 1 requires the researcher to commit to a hypothesis in writing before opening a single data file.

1.1 Theory and the Source of Edge

Legitimate sources of trading edge fall into a small number of categories. Structural edges arise from market microstructure — the mechanics of order matching, liquidity provision, and information incorporation. Behavioral edges arise from systematic cognitive biases in market participants: anchoring, herding, overreaction, and loss aversion all create predictable mispricing's. Risk premium edges arise from compensation demanded for bearing specific risks — momentum, value, carry, and low-volatility are well-documented examples. The researcher must identify which category their hypothesis falls into and articulate the mechanism with enough precision to generate a testable prediction. A hypothesis that cannot specify conditions under which it should fail is not a scientific hypothesis — it is a rationalization.

Academic Reference

Lopez de Prado (2018) argues that most practitioner-backtested strategies are the result of data mining rather than genuine research. He estimates the bar for statistical significance must account for the number of implicit tests conducted during the research process.

1.2 Defining the Market Inefficiency

The inefficiency must be defined with sufficient precision to specify: the observable signal indicating it is present; the expected direction and magnitude of subsequent price movement; the timeframe over which it resolves; and the population of instruments in which it should be observable. A well-defined inefficiency also predicts where it should not exist — these boundary conditions are evidence of genuine predictive content rather than a vague assertion.

1.3 Choosing Strategy Paradigm: Rule-Based, ML, or Hybrid

Rule-based strategies are optimal when the inefficiency can be characterized by a small number of conditions the researcher can specify explicitly. The signal is interpretable, the parameter space is small, and overfitting risk is manageable. ML strategies are optimal when the inefficiency involves complex non-linear interactions between many variables — patterns no human researcher would explicitly specify but that exist consistently in data. Hybrid strategies — the most common structure in professional systematic trading — use rule-based

logic for well-understood components and ML where conditional adaptive behavior adds genuine value: trade filtering, dynamic sizing, and exit timing.

Rule-Based Track	ML Track
Entry, exit, sizing, and regime filter are all explicitly specified by the researcher. Parameters are tuned but logic structure is fixed.	A model learns signal relationships from data. Researcher specifies the feature set, target variable, and model architecture — not the rules themselves.

1.4 Instrument, Timeframe & Regime Scope

Instrument and timeframe selection are hypothesis-driven decisions. Timeframe follows from the expected duration of the inefficiency — a hypothesis based on institutional order flow accumulation implies a multi-day signal horizon; a microstructure hypothesis implies intraday resolution. Regime scope defines which market conditions the strategy is designed for, preventing post-hoc rationalization of sub-period underperformance.

1.5 Pre-Specifying Performance Thresholds

Minimum performance criteria must be documented before any backtest is run: annualized Sharpe Ratio above 1.0 net of costs; maximum drawdown within risk tolerance; minimum 50 trades (100+ preferred); Profit Factor above 1.5. For ML strategies, additional thresholds include minimum meta-label precision/recall and maximum acceptable accuracy degradation between IS and OOS. Writing these down before testing prevents goal-post shifting after results are seen.

1.6 The Multiple Comparisons Problem

Bailey, Borwein, Lopez de Prado, and Zhu (2014) formalized the multiple comparisons problem with the Deflated Sharpe Ratio, which adjusts observed Sharpe downward based on the number of trials conducted. A strategy with observed Sharpe of 1.5 found after testing 50 variants may have a Deflated Sharpe statistically indistinguishable from zero. Every variant tested — even informally — counts toward the tally and must be tracked. Multiple testing corrections such as Bonferroni or the Deflated Sharpe Ratio should be applied whenever more than one configuration is evaluated on the same dataset.

Academic Reference

Bailey et al. (2014), 'The Probability of Backtest Overfitting', demonstrates that the probability of selecting an overfit strategy increases dramatically with the number of trials. Their Combinatorially Symmetric Cross-Validation (CSCV) framework provides a practical method for estimating this risk.

PHASE 2: Data Infrastructure

Constructing a clean, unbiased dataset, IS/OOS split, and the ML feature matrix

The quality of a strategy's backtest is bounded absolutely by the quality of its data. Phase 2 addresses the entire pipeline from acquisition through cleaning, bias elimination, and the IS/OOS split. For ML strategies it extends into feature matrix construction, labelling, and sample weighting — three problems unique to financial ML that, if mishandled, produce models that appear to work in backtesting but fail completely in live trading.

2.1 Data Sourcing & Vendor Selection

Historical data falls into several categories: price and volume (OHLCV bars), tick and order book, fundamental, alternative (sentiment, satellite, credit card), and reference data. Critical vendor considerations include adjustment for corporate actions, inclusion of delisted securities, and data resolution relative to the strategy's timeframe. For ML strategies, data quantity is an additional constraint — models require a minimum number of labelled examples to train reliably, which dictates the minimum historical depth required.

2.2 Data Cleaning & Integrity Checks

Raw data invariably contains errors: price spikes from erroneous prints; OHLC relationship violations; zero or negative prices or volumes; missing bars; duplicate rows; and timezone errors around daylight saving transitions. Each type of error produces specific backtest distortions. The cleaning process must be systematic, documented, and applied before any feature construction or signal development begins. Every modification to raw data should be logged with its justification.

2.3 Survivorship Bias & Point-in-Time Data

Survivorship bias — testing only on assets that currently exist while ignoring those that were delisted, went bankrupt, or were acquired — inflates historical returns by 1-5% annually in equity strategies. The solution is a point-in-time database that records universe composition as it was known on each historical date. For fundamental data, point-in-time integrity requires using values as reported on their release date, not as subsequently restated — a distinction that affects earnings-based strategies substantially.

2.4 The IS / OOS Split

The chronological division into an in-sample (IS) development set and a sealed out-of-sample (OOS) validation set is the most consequential structural decision in the workflow. OOS must be the most recent period, must be split before any data exploration begins, and must remain physically inaccessible until Phase 6. Common ratios are 60/40 to 80/20 IS/OOS. OOS must contain enough trades for statistical significance — minimum 30, preferably 50+. Even viewing summary statistics of the OOS period constitutes a form of information leakage.

IS / OOS Data Split and Walk-Forward Windows

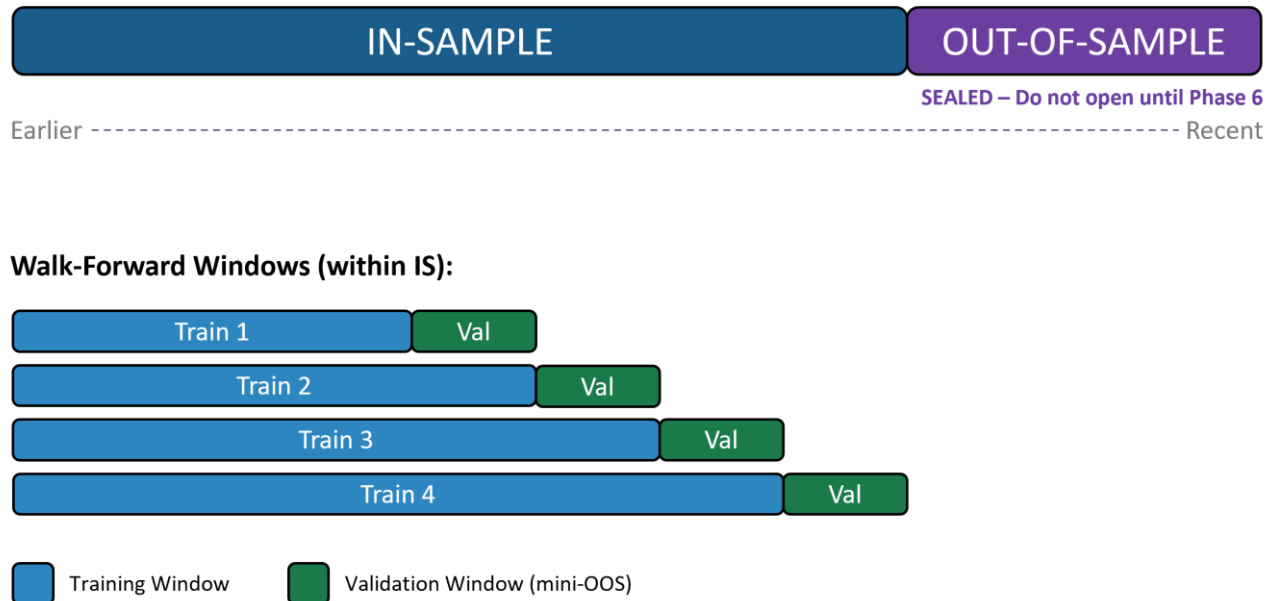


Figure 2. Chronological IS/OOS split. The IS period contains all development work including nested walk-forward windows. The OOS period is sealed until Phase 6 and opened exactly once.

2.5 Look-Ahead Bias at the Data Layer

All signals at bar N must use only data available through bar N-1, with entries at bar N+1 open at earliest. For ML strategies, preprocessing objects must be fit exclusively on IS training data — fitting a StandardScaler on the full dataset before splitting contaminates every validation fold with future distribution information. This lagging rule must be enforced mechanically in the backtesting framework, not left to the researcher's judgement.

2.6 Feature Matrix Construction (ML)

The feature matrix is a two-dimensional array where each row is a trade decision point and each column is a feature. Financial observations violate two standard ML assumptions: serial correlation means they are not independent, and shifting volatility regimes mean they are not identically distributed. Lopez de Prado (2018) notes that a dataset with 1,000 daily bars may contain only 50-100 truly independent observations. Features must be stationary — typically requiring transformations such as returns rather than prices, z-scores relative to rolling windows, or fractional differencing. All features must be lagged to prevent look-ahead bias.

ML-Specific

Lopez de Prado (2018) Chapter 4 covers fractional differencing — a technique that achieves stationarity while preserving long-range memory in the series, improving on simple differencing which eliminates the correlations that make prices predictable.

2.7 Labelling: The Triple-Barrier Method (ML)

Labelling assigns a target outcome to each row of the feature matrix. The naive approach of labelling with the sign of the next N-bar return has problems: an arbitrary horizon, no risk accounting, and overlapping windows that create serial correlation in labels. The triple-barrier method addresses these by placing three barriers for each observation: an upper barrier at a profit target level (e.g., 2% above entry), a lower barrier at a stop-loss level (e.g., 1% below entry), and a vertical barrier at a maximum holding period. The label is determined by which barrier is touched first: upper hit produces label +1; lower hit produces label -1; vertical hit without a horizontal barrier produces label 0. Barrier levels are set based on the strategy's intended stop-loss and profit target rules from Phase 3, creating a direct connection between ML labelling and live trade management.

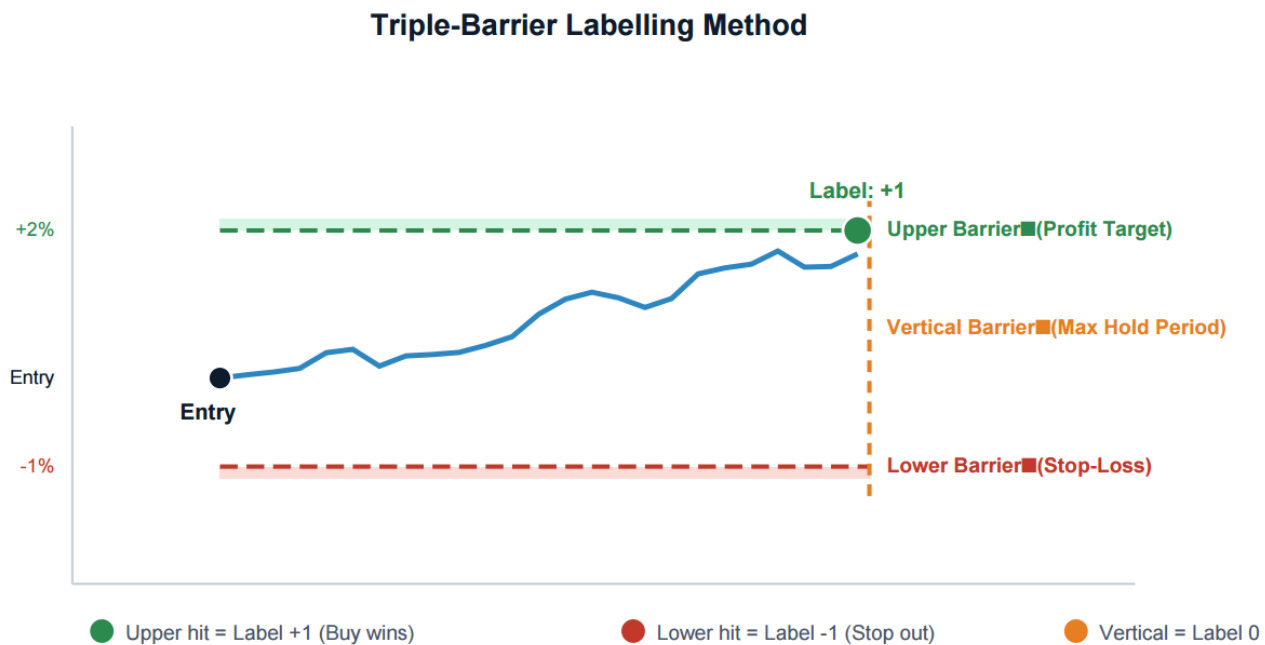


Figure 3. Triple-barrier labelling. The price path hits the upper barrier first, generating label +1. Had it hit the lower barrier first the label would be -1. The vertical barrier prevents labels based on indefinite holding.

Academic Reference

Lopez de Prado (2018) Chapter 3 provides the full mathematical treatment of the triple-barrier method. The method is also the foundation for the meta-labelling framework described in Phase 3.

2.8 Sample Weighting for Overlapping Returns (ML)

When observations have overlapping label windows, consecutive observations are not independent. Uniqueness weighting assigns each observation a weight proportional to the fraction of its label window that does not overlap with adjacent observations — observations with more unique information content receive higher weights. Time-decay weighting additionally discounts older observations, reflecting that more recent data is more representative of the current market environment. Both approaches are implemented as sample weights passed to the model's training routine.

PHASE 3: Strategy Design & Initial Backtesting

Defining all strategy components before the first backtest is run

All components — signal logic, ML pipeline, position sizing, stop-losses, and cost assumptions — must be fully specified before the first backtest. This is a hard requirement. A backtest without defined position sizing produces meaningless P&L. One without stop-loss rules produces an unrealistic drawdown profile. For ML strategies, the pipeline architecture and labelling methodology must be defined before training begins — modifying these after seeing results is data mining. Chan (2009) makes this explicit: the complete strategy specification must be committed to before backtesting.

3.1 Signal Logic & Feature Engineering

For rule-based strategies, signal logic is the explicit conditional specification of when to enter, exit, and in which direction. For ML strategies, it is the feature set plus the target variable — the model learns the conditional relationships from data. Effective features fall into trend (moving averages, breakouts), momentum (rate of change, RSI), volatility (ATR, Bollinger Band width), volume (volume ratio, OBV), microstructure (bid-ask spread, order imbalance), and cross-sectional (ranking relative to peers) categories. The feature set should be the minimum that adequately captures the hypothesized signal.

3.2 ML Pipeline Architecture (ML)

The ML pipeline is the ordered sequence of data transformations and model operations that converts raw features into trade signals. Standard stages: (1) feature computation with lags applied; (2) preprocessing — scaling, encoding, imputation — fit on IS training fold only and applied without refitting to validation and OOS; (3) feature selection; (4) the model itself. The pipeline must be encapsulated as a single object so that fitting the pipeline on training data and applying it to validation data enforces the leakage boundary automatically.

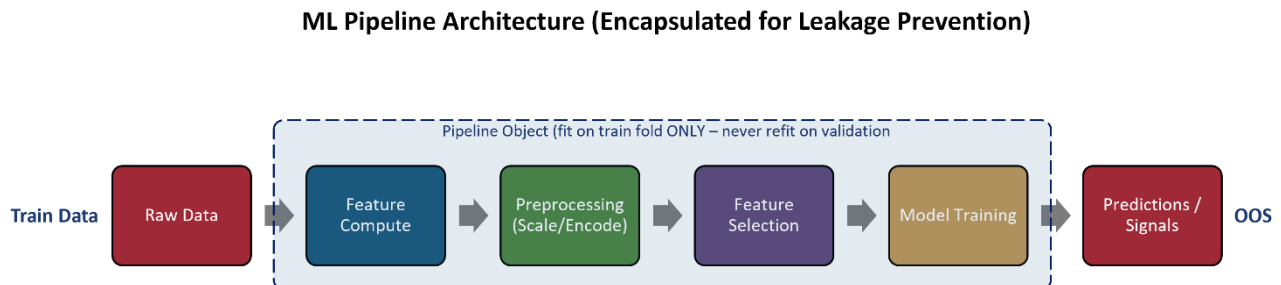


Figure 4. ML pipeline architecture. All preprocessing steps must be inside the pipeline object and fit only on the training fold. The enclosure boundary marks where fit-only-on-training-fold applies. Preprocessing is applied — never refit — to validation and OOS data.

ML-Specific

A critical rule: the pipeline must be fit as a complete unit on each training fold during cross-validation. Fitting preprocessing on the full IS dataset before splitting into folds introduces look-ahead bias into every validation fold. All preprocessing steps — scalers, encoders, imputers, feature selectors — must be inside the pipeline object.

3.3 Meta-Labeling: The Secondary Model Framework (ML)

Meta-labeling separates the task of determining trade direction from the task of determining whether to take a trade. A primary model — which may be rule-based or ML — generates candidate signals with a direction. A secondary meta-label model evaluates each candidate and predicts whether it will be profitable, producing a binary filter label (take or skip) and a confidence score (size proportionally). The meta-label model trains on all historical instances of the primary signal firing, labelled via triple-barrier, with features describing market conditions at each instance: volatility regime, trend strength, volume profile, time of day, spread width, and recent win rate of the primary signal.

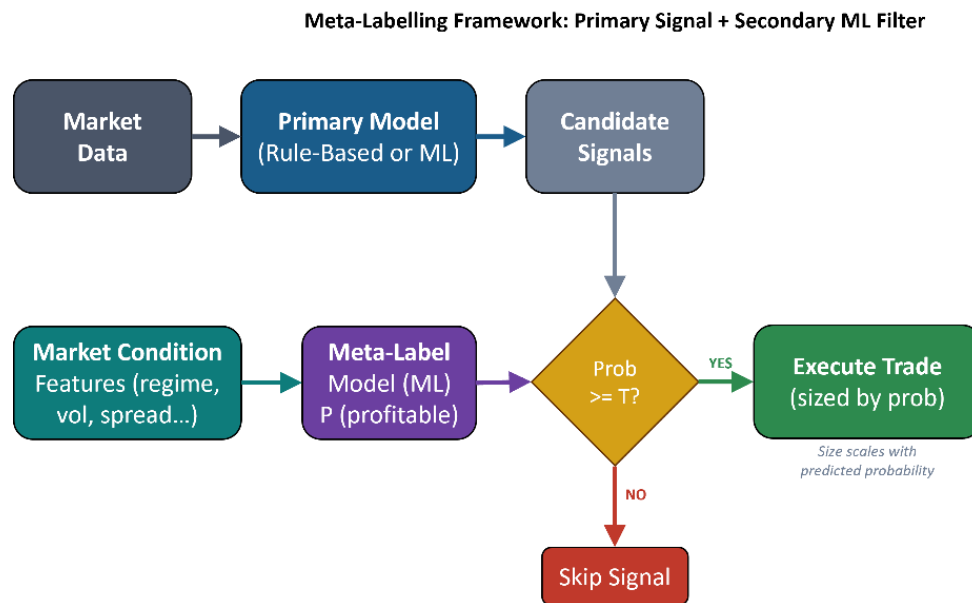


Figure 5. Meta-labelling framework. The primary model generates candidate signals; market condition features feed the meta-label model; its probability output determines both whether to take the trade and how large to size it.

Academic Reference

Lopez de Prado (2018) Chapter 3 introduces meta-labelling as a solution to training ML models on financial data with low signal-to-noise ratios. By first using a primary model to filter the universe of observations to those near trade decisions, the meta-label model trains on a more informative subset.

3.4 ML-Based Trade Filter (ML)

The trade filter is a binary classification model trained to distinguish profitable from unprofitable instances of the primary signal. Positive training examples are primary signal instances where the upper triple-barrier was hit first. Negative examples are stop-outs or expired vertical barrier trades. The probability threshold is a hyperparameter tuned in Phase 4 — it represents a trade-off between precision (proportion of taken trades

that are profitable) and recall (proportion of profitable trades that are taken). A higher threshold increases precision but reduces the number of trades taken.

Rule-Based Track	ML Track
Filtering via explicit regime rules: 'only trade when VIX is below 25 and price is above the 200-day MA'. Researcher specifies conditions based on theory.	Filtering via trained binary classifier that evaluates a feature vector of current conditions and outputs a probability. Learns filtering conditions from historical trade outcomes.

3.5 Position Sizing

Position sizing must be defined before the first backtest because it directly determines P&L distribution, drawdown profile, and all subsequent risk metrics. Fixed fractional sizing risks a fixed percentage of equity per trade (1-2% typical). Volatility targeting sizes each position so that a one-ATR adverse move costs a fixed equity percentage, keeping risk consistent across changing volatility regimes. The Kelly Criterion provides the theoretically optimal fraction but is typically implemented at half- or quarter-Kelly to manage drawdown variance. For portfolio strategies, maximum single-position weight, sector concentration limits, and total gross exposure limits must also be defined upfront.

3.6 Dynamic Position Sizing via ML (ML)

The meta-label model's predicted probability maps directly to a position size multiplier. At maximum confidence (probability near 1.0), the full base position size is deployed. At minimum confidence near the threshold, a fraction of base size is deployed. Between these extremes, size scales proportionally. This concentrates capital in the highest-confidence trades and reduces exposure on marginal ones — improving the risk-adjusted return of the strategy without changing the entry signal itself.

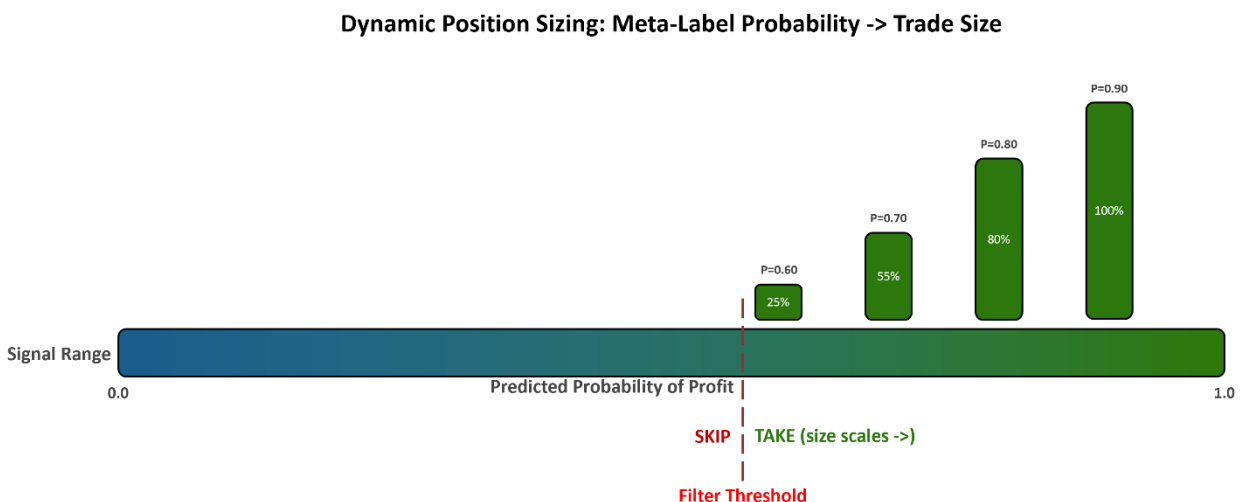


Figure 10. Dynamic position sizing. Signals below the filter threshold are skipped entirely. Above it, position size scales proportionally with the meta-label model's predicted probability of profit.

Rule-Based Track	ML Track
Fixed function of equity, volatility, or Kelly fraction. Does not vary with signal confidence beyond the volatility normalization already in the formula.	Scales with meta-label model's predicted probability. High-confidence trades receive larger allocations; near-threshold trades receive smaller allocations or are skipped entirely.

3.7 Stop-Loss & Exit Rules

Stop-loss and exit rules define when positions are closed and determine the distribution of trade returns. They must be defined before backtesting because they directly determine win rate, average win, average loss, and Profit Factor — which for ML strategies also determine the triple-barrier labelling levels. ATR-based stops normalize stop distance to current volatility. Trailing stops capture extended moves by advancing the stop in the direction of the trade. Scaled exits take partial profits at predetermined levels while trailing the remainder. Time-based stops close after a maximum holding period. Exit structure must be consistent with the hypothesis about how prices move after entry.

3.8 Transaction Cost Modelling

Every backtest must apply realistic transaction costs to every trade. The three components are: commission (the broker's explicit charge); bid-ask spread (the cost of crossing the spread on each side); and slippage (additional execution disadvantage from market impact, latency, and adverse selection). For liquid large-cap instruments, total round-trip costs are typically 2-10 basis points. Cost assumptions should be pessimistic — an overly optimistic model produces results that cannot be replicated in live trading, which is the more dangerous error.

3.9 Capacity Constraints

Every strategy has a maximum capital deployment level beyond which its own trading moves prices against itself, eroding the edge. A strategy's intended position size should not exceed 1-5% of average daily volume (ADV) in any instrument. Capacity analysis must be completed before the backtest is considered valid because it defines the deployment scale at which the results are meaningful. A strategy viable at \$1M may be completely unviable at \$100M due to market impact costs not captured in the backtest.

3.10 Running the First Backtest

With all components defined — signal logic, ML pipeline if applicable, position sizing, stop-losses, cost assumptions, and capacity constraints — the initial IS backtest is run. This first run is diagnostic: it verifies the implementation is correct, confirms the trade count is sufficient for statistical testing, checks for implausible results suggesting data or logic errors, and establishes the baseline performance against which optimization in Phase 4 will be measured.

PHASE 4: In-Sample Optimization

Systematically improving strategy parameters within IS data without overfitting

Optimization searches the parameter space for configurations that improve generalized performance. It is a necessary step but the primary mechanism through which overfitting is introduced. The discipline of Phase 4 is to optimize in ways that improve genuine out-of-sample performance rather than IS-specific fit.

4.1 Hyperparameter Tuning

For rule-based strategies: indicator lookback periods, threshold levels, timing parameters. For ML models: tree depth, number of estimators, regularization coefficient, learning rate. The goal is parameter settings that generalize to unseen data, not that fit the IS sample perfectly. Overfitting signatures include narrow optimal parameter regions where performance degrades sharply with small changes, excellent IS performance that collapses in OOS, and parameters that lack theoretical motivation.

4.2 Time-Series Cross-Validation

Standard k-fold cross-validation randomly assigns observations to folds, which for time-series data allows future information to appear in training sets — introducing look-ahead bias. Time-series CV enforces chronological ordering: the validation set always follows the training set in time. The expanding window scheme grows the training set with each fold; the rolling window scheme advances both windows together. Performance should be evaluated as the average across all validation folds, with the standard deviation as a stability measure.

4.3 ML Pipeline CV & Leakage Prevention (ML)

The pipeline must be treated as a single unit during cross-validation. Every preprocessing step must be fit on the training fold and applied (not refit) to the validation fold. All steps must be inside the pipeline object before calling the CV routine. A useful diagnostic: randomly shuffle the target variable in the training set and check whether CV performance drops to chance level. If the model retains significant predictive performance with a shuffled target, some form of leakage is present.

ML-Specific

Fitting a scaler on the full IS dataset before cross-validation is the single most common pipeline leakage error in financial ML. It contaminates every validation fold with mean and variance information from the validation period itself, making the model appear more accurate than it actually is.

4.4 Feature Selection

Correlation-based filtering removes features with pairwise correlation above 0.85. Mutual information scores measure non-linear dependence between each feature and the target. SHAP (SHapley Additive exPlanations) values measure each feature's marginal contribution to predictions in a trained model, providing reliable

importance estimates that account for feature interactions. The principle of parsimony applies: use the minimum number of features that adequately captures the signal. Features without theoretical motivation should be excluded regardless of apparent IS importance.

4.5 Meta-Label Model Optimization (ML)

The meta-label model requires its own optimization within IS using TimeSeriesSplit cross-validation. Key hyperparameters include the probability threshold, model architecture, regularization settings, and feature set. The evaluation metric should reflect the trading objective rather than generic classification accuracy — F1 score on the positive class is more informative than overall accuracy when classes are imbalanced. The threshold selected during optimization should be the one that maximizes the risk-adjusted return of the combined strategy on CV folds, not the one that maximizes the meta-label model's classification metrics in isolation.

4.6 Avoiding Overfitting During Optimization

L1 regularization drives some coefficients to zero performing feature selection; L2 regularization shrinks all coefficients toward zero reducing sensitivity to individual features. Constraining parameter search to theoretically motivated ranges prevents nonsensical historical fits. Lopez de Prado (2018) recommends selecting the configuration with the best risk-adjusted performance across the full walk-forward CV — systematically preferring robustness over peak performance.

PHASE 5: Robustness & Stress Testing

Subjecting the fully specified strategy to adversarial conditions before OOS validation

Stress testing is conducted entirely within the IS dataset. Every test functions as a gate — strategies that fail are returned to Phase 3 for redesign. Only strategies that pass all applicable tests advance to OOS validation. For ML strategies, additional model-specific tests assess whether the pipeline's predictive properties are stable across sub-periods and distributional shifts.

5.1 Parameter Sensitivity Analysis

Parameter sensitivity examines how performance changes as optimized parameters move away from their optimal values. Robust strategies show smooth, gradual degradation — broad plateaus rather than sharp peaks. Each parameter is perturbed independently across a range of $\pm 10\%$, $\pm 20\%$, and $\pm 30\%$ from its optimal value. For ML strategies, this extends to model hyperparameters. A model whose performance collapses with a 10% change in regularization strength was fitting noise, not learning a stable signal.

Parameter Sensitivity: Robust Plateau vs Overfit Peak

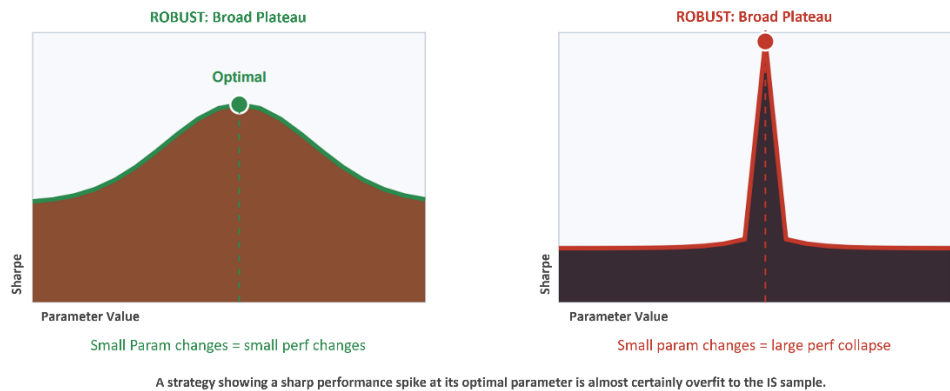


Figure 8. Parameter sensitivity landscapes. Left: a robust strategy shows a broad plateau — performance degrades gradually from the optimal. Right: an overfit strategy shows a sharp spike — performance collapses immediately when parameters change slightly.

5.2 Noise Injection & Cost Sensitivity

Noise injection adds random perturbations to entry and exit prices across thousands of simulation runs, producing a distribution of outcomes that reflects realistic execution uncertainty. Cost sensitivity analysis varies the transaction cost assumption from zero to a multiple of the base estimate, identifying the break-even cost level at which the edge is consumed. A strategy that requires transaction costs below 1 basis point to be profitable is fragile — any deviation from perfect execution destroys the edge.

5.3 Monte Carlo Simulation

Monte Carlo simulation generates an empirical distribution of outcomes by randomly resampling the strategy's trade return sequence thousands of times and reconstructing an equity curve from each shuffled sequence. Primary outputs are the distribution of drawdowns — particularly the 5th percentile worst-case — and the probability of ruin below a defined threshold. These are direct inputs to position sizing decisions: if the worst-case drawdown distribution is unacceptable at base position size, size must be reduced before deployment.

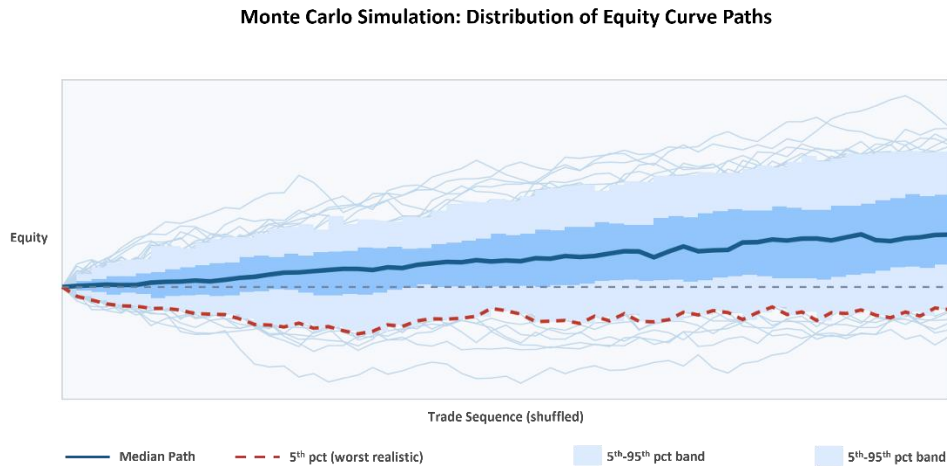


Figure 7. Monte Carlo distribution of equity curve paths. The median path represents expected performance; the 5th percentile path represents the realistic worst case. Shaded bands show the interquartile and 5th-95th percentile confidence intervals.

5.4 Sub-Period Variance Analysis

The IS period is divided into equal sub-periods and strategy performance is evaluated independently within each. A robust strategy should produce consistently positive results across the majority of sub-periods — not derive all aggregate IS performance from one exceptional window. High sub-period variance indicates regime dependence and directly informs the regime analysis below.

5.5 Walk-Forward Analysis

Walk-forward analysis simulates the real-world process of developing, deploying, and periodically reoptimizing a strategy over time. A training window is used to optimize the strategy; the optimized parameters are applied to an immediately following validation window; both windows advance forward by one validation length; the process repeats until the end of IS data. The aggregated validation performance across all windows represents realistic performance under periodic recalibration. Pardo (2008) established this methodology as the standard robustness test for systematic trading strategies.

Walk-Forward Analysis: Rolling Optimize and Validate Windows

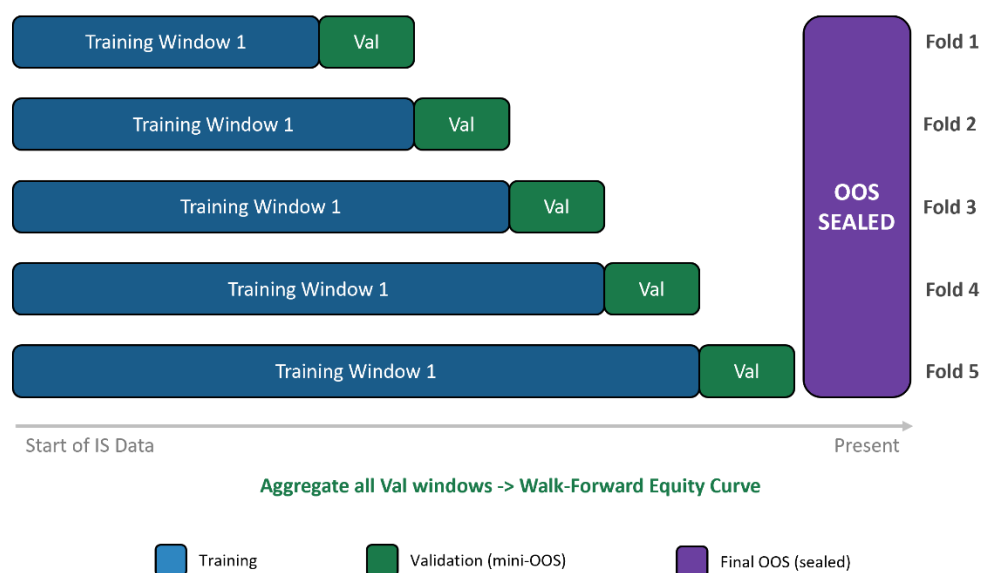


Figure 6. Walk-forward analysis across five folds. Training windows expand with each fold; validation windows immediately follow. The final OOS block remains sealed throughout. All validation results are aggregated into a single walk-forward equity curve.

5.6 Regime Analysis

Regime analysis segments the IS period by market condition and evaluates strategy performance within each independently. Regimes are defined exogenously using observable variables (200-day MA for bull/bear, VIX for volatility, interest rate levels for macro regime) or endogenously via Hidden Markov Models or k-means clustering on return characteristics. Understanding the regime profile explains sub-period variance and informs the design of regime filters that can disable the strategy in adverse conditions.

5.7 ML-Specific Robustness Tests (ML)

Feature importance stability measures whether the model consistently relies on the same features across IS sub-periods and walk-forward windows. SHAP importance distributions should be consistent — a model using different top features in each sub-period is fitting sub-period specific noise rather than a stable signal. Distributional shift testing evaluates model performance when the input feature distribution changes. The label leakage audit shuffles target labels and verifies that performance drops to chance level — if it does not, the pipeline contains leakage.

ML-Specific

A model that passes all standard stress tests but fails the label leakage audit is one of the most dangerous outcomes in ML strategy development. It will show excellent backtest results then fail completely in live trading when future information is not available.

5.8 Factor Decomposition

Regressing strategy returns against Fama-French factor returns (market, size, value, momentum, profitability, investment) isolates the alpha component. The intercept (Jensen's alpha) represents return not explained by

factor exposures. A statistically significant positive alpha after controlling for standard factors is evidence of genuine edge. High R-squared with near-zero alpha indicates the strategy is a disguised factor product. This test applies equally to rule-based and ML strategies — an ML model can learn to mimic factor exposures as easily as it can learn genuine alpha.

5.9 Multiple Testing Correction

The Bonferroni correction divides the desired significance level by the number of tests. The Deflated Sharpe Ratio adjusts the observed Sharpe downward based on the number of trials conducted and the statistical properties of the return series. A Deflated Sharpe Ratio below zero indicates that the strategy's observed performance is statistically indistinguishable from a lucky draw from the space of tested strategies. For ML strategies, the number of implicit tests is typically larger due to the greater number of architectural choices made during pipeline design.

PHASE 6: Out-of-Sample Validation

A single, unmodified evaluation on data the strategy has never influenced

OOS validation is the most critical step in the workflow. It is the only genuinely independent test of whether IS performance reflects a persistent edge or an overfit artefact. For ML strategies, this means applying the fully trained pipeline — with preprocessing fit on IS data, model weights fixed from IS training, and meta-label threshold set during IS optimization — to the OOS dataset exactly once.

6.1 The One-Shot Rule

OOS must be run exactly once with all parameters and pipeline components frozen precisely as they were at the conclusion of Phase 5. No adjustments may be made after OOS results are observed. Each time a strategy is modified in response to OOS results and re-evaluated, the OOS dataset effectively becomes part of the training data. After two or three such iterations, the OOS period is as contaminated as the IS period. For ML strategies, the scaler, feature selector, model weights, and meta-label threshold are all frozen before OOS is unsealed.

6.2 Interpreting Performance Degradation

Some performance degradation from IS to OOS is expected and healthy. A rough rule of thumb consistent with Chan (2013) is that OOS Sharpe Ratio should be at least 50% of IS Sharpe Ratio. For ML strategies, additional degradation is expected due to distributional shift between the training and OOS periods. OOS Sharpe below 30% of IS Sharpe, or negative OOS Sharpe where IS was strongly positive, reliably indicates overfitting regardless of strategy type.

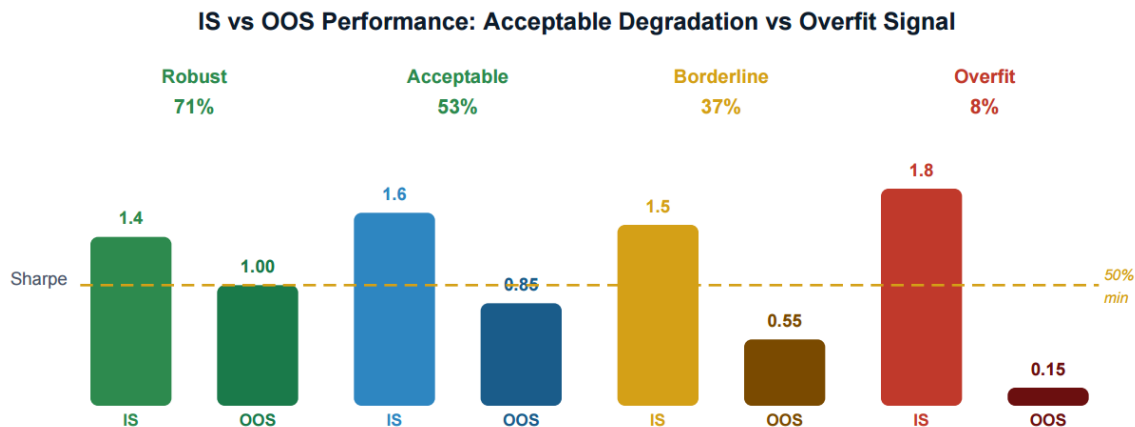


Figure 9. IS vs OOS Sharpe Ratio across four outcome categories. The percentage shown is the OOS/IS ratio. The dashed line marks the 50% minimum threshold. An OOS/IS ratio below 30% is the signature of an overfit strategy.

6.3 Sub-Period Consistency within OOS

The OOS period should be divided into sub-periods to verify that aggregate performance is not driven by a single favorable window. For ML strategies, sub-period OOS performance should be compared against the regime profile established in Phase 5 — if OOS happened to contain the strategy's best regime, aggregate OOS results will be optimistic relative to what should be expected across all regimes. Consistent positive performance across OOS sub-periods spanning different market conditions is the strongest evidence of genuine generalizability.

6.4 Handling OOS Failure

A failed OOS test is definitive. The correct response is to close the research iteration, document the outcome fully, and return to Phase 1 with a new hypothesis. Specific temptations to resist: parameter tweaking based on OOS results, which converts OOS into training data; period rationalization, arguing the OOS regime was unusual; threshold revision, lowering pre-specified performance criteria; and for ML strategies, retraining on a combined IS+OOS dataset, which eliminates the independence of the test entirely.

Important

The discipline of accepting OOS failure and returning to Phase 1 is in practice the hardest part of the entire workflow. The sunk cost of ML pipeline development creates particularly strong pressure to rationalize continued iteration. Researchers who maintain this discipline are those whose live results most closely match their backtested and OOS results.

PHASE 7: Paper Trading & Forward Validation

Validating live execution mechanics before real capital is deployed

Paper trading bridges the gap between backtesting and live deployment by running the strategy in real-time market conditions with simulated order execution. For ML strategies, paper trading additionally validates that the live feature computation pipeline produces inputs matching the training distribution — a critical check because subtle differences in real-time versus historical feature computation can degrade model performance substantially.

7.1 Purpose & Setup

The primary purpose is to detect discrepancies between backtest behavior and live behavior: differences in bar construction between live and historical data feeds; latency between signal generation and order submission; edge cases in live data such as trading halts and auction periods; and for ML strategies, differences in the feature distribution between real-time and historical computation. Paper trading must use identical code, data feed, and broker integration as live deployment — not a simplified simulation.

7.2 Implementation Shortfall

Implementation shortfall — formally defined by Perold (1988) — is the aggregate difference between theoretical and actual execution performance. Its components are: delay cost (price movement between signal generation and order submission); spread cost (cost of crossing the bid-ask spread); market impact (adverse price movement from the act of trading); and timing risk (price movement while an order is being worked). Measuring these during paper trading calibrates the transaction cost model and identifies execution bottlenecks before real capital is at risk.

7.3 Statistical Significance of Forward Results

Short paper trading periods — two weeks, one month — are insufficient to draw meaningful conclusions. A minimum of 30-50 trades is required; 100+ for reasonable confidence. The paper trading period should span diverse market conditions. For ML strategies, paper trading also validates whether the meta-label model's probability predictions are well-calibrated — whether trades assigned 70% probability actually win approximately 70% of the time.

PHASE 8: Deployment & Lifecycle Management

Going live and maintaining strategy and model health over the operational lifetime

Deployment is the beginning of the strategy's operational lifecycle. Live performance monitoring, decay detection, and the discipline of scheduled rather than reactive retraining are as important to long-term success as the development phases that preceded them. For ML strategies, model drift monitoring adds an additional layer of operational responsibility.

8.1 Pre-Deployment Checklist

Before committing real capital: confirm live data feed signals match paper trading signals on the same data; verify order routing is connected to the correct account with correct sizing parameters; confirm all risk limits are active in the execution system; test that the kill switch halts all activity within seconds; verify monitoring and alerting infrastructure is live. For ML strategies: confirm live feature computation produces inputs with the same statistical properties as training data; verify the meta-label model loads the correct trained weights; confirm position sizing reads model probability outputs correctly. Deploy at 25-50% of intended full size initially, scaling up gradually as operational confidence develops.

8.2 Performance Monitoring Framework

Track metrics at multiple time horizons. Daily: P&L, trade count, fill quality relative to signal price, risk limit utilization. Weekly: rolling 90-day Sharpe, win rate, average win/loss ratio versus IS and OOS benchmarks. Monthly: cumulative performance since inception, drawdown from peak, and regime attribution. Alert thresholds should be pre-specified from Monte Carlo outputs in Phase 5: a 5th-percentile drawdown breach triggers a review; sustained sub-threshold Sharpe for a defined number of days triggers a strategy review.

8.3 Strategy Decay & Edge Erosion

All trading edges decay over time as market structure changes and other participants discover the same inefficiency. The Chow test and CUSUM test identify structural breaks in the return series where statistical properties appear to have shifted. A sustained decline in rolling Sharpe that persists across multiple market regimes and is not explained by known adverse conditions is the clearest practical indicator of edge erosion. When decay is confirmed, review the original hypothesis: has the mechanism that created the edge changed? If yes, retire the strategy rather than modify it without theoretical justification.

8.4 Model Drift Detection (ML)

ML models face model drift — the relationship between input features and the target variable changing over time — beyond general strategy decay. Covariate shift (the input distribution changing) is monitored via Population Stability Index (PSI) and Kullback-Leibler divergence on individual features. When PSI exceeds 0.2 for a significant feature, a retraining review should be triggered. Concept drift (the feature-outcome relationship

changing) is monitored via the meta-label model's calibration in live trading — a systematic decline in win rate for high-confidence trades not explained by market regime is evidence of concept drift.

ML-Specific

Model drift is distinct from strategy decay. A strategy can have a persistent underlying edge while its ML model's learned representation of that edge becomes stale. Monitoring both strategy-level performance metrics and model-level drift metrics is required to correctly diagnose live performance degradation in ML strategies.

8.5 Retraining Discipline

Retraining must be governed by pre-specified schedules, not by observed performance. Reactive retraining triggered by poor performance is statistically equivalent to the IS overfitting the workflow was designed to prevent. Scheduled retraining at fixed calendar intervals is the operationally sound alternative. Each retraining event is a mini-repetition of the full development workflow: fresh IS/OOS split on the updated dataset, preoptimization using the same methodology, and validation before deploying new parameters. For ML strategies this means relabeling, sample reweighting, cross-validated hyperparameter search, and meta-label threshold preoptimization. All retraining events must be documented with full records of the data used, parameters before and after, and validation results.

Academic Reference

Chan (2013) emphasizes that operational discipline in live deployment is as important as development rigor. Strategies managed reactively in production tend to converge toward poor risk-adjusted returns as continuous adjustments in response to recent performance replicate the IS overfitting problem in real time.

Phase Sequence Reference

The table below summarizes all eight phases, their key universal tasks, ML-specific additions, and the gate condition that must be satisfied before advancing. The sequencing reflects logical dependencies — no phase may be skipped or reordered without invalidating the statistical integrity of the workflow.

Ph	Phase	Universal Tasks	ML-Specific Additions	Gate Condition
1	Research & Hypothesis	Hypothesis, mechanism, paradigm selection, thresholds, multiple comparisons register	Specify ML/rule-based/hybrid; define labelling target type	Documented hypothesis; thresholds set; no data examined
2	Data Infrastructure	Source, clean, survivorship bias, point-in-time, IS/OOS split, look-ahead audit	Feature matrix; triple-barrier labelling; sample weighting	OOS sealed; pipeline audited; labels and weights computed
3	Strategy Design	Signal logic, sizing, stop-losses, cost model, capacity, first backtest	ML pipeline architecture; meta-labelling; trade filter; dynamic sizing	All components defined before first backtest; pipeline encapsulated
4	IS Optimization	Hyperparameter tuning via TimeSeriesSplit CV; feature selection	Pipeline CV with leakage prevention; meta-label threshold optimisation	Optimization via CV only; no leakage; no OOS touched
5	Stress Testing	Param sensitivity, noise, Monte Carlo, sub-period, walk-forward, regime, factor decomp, multiple testing	Feature importance stability; distributional shift; label leakage audit	All tests passed; Deflated Sharpe acceptable; no leakage detected
6	OOS Validation	Single frozen OOS run; degradation analysis; sub-period consistency	Frozen pipeline weights and meta-label threshold; calibration check	OOS Sharpe $\geq$ 50% IS; meets pre-specified thresholds; no modifications
7	Paper Trading	Live signal generation; implementation shortfall; statistical significance	Feature distribution match vs training; meta-label probability calibration	Live performance consistent; 30+ trades; feature distributions stable
8	Deploy & Monitor	Pre-deployment checklist; performance monitoring; decay detection; scheduled retraining	Model drift monitoring (PSI, KL); probability recalibration	Continuous — monitoring and scheduled retraining never stop

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